

Big Titles for \$400

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Abstract

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1 Introduction

The Laurentian Great Lakes contain a fifth of Earth’s available freshwater and as a result, the climate of this region is unlike any other on the planet. The lakes act to thermally regulate the surrounding land by means of absorbing and releasing heat as synoptic weather systems propagate through the region. Depending on the season and position of the jet stream, the prevailing Westerlies bring warm air from the southwestern US or cold air from northern Canada. Strong winds in the fall act to mix the water column of the lakes and also increase evaporation which contributes further to surface cooling. These interactions between lake surface and atmosphere play out year-round and have significant influence on the lives of those who call the Great Lakes home.

Lake ice is extremely variable in the Great lakes but has impacts ranging from economy to ecology. For example, the commercial shipping industry in the Great Lakes is estimated to support nearly a quarter million jobs, transporting 135 million tons of cargo and generating \$6 billion in tax revenue annually (Associates, 2023). The presence vs absence and timing of ice cover influences the length of the shipping season. Essential goods are

is an unique phenomena, in the sense that unlike temperature , wind speeds or water currents, it’s bounded by 0 and 1 and it either is present or it’s not

Previous studies have demonstrated that, consistent with trends in air and ocean temperatures, lake temperatures—including those of the Great Lakes, are also increasing (Austin and Colman, 2007,) However, the rate and extent of warming in the Great Lakes varies considerably from lake to lake and even *within* each lake, partly due to the heterogeneity of characteristics within lakes (bathymetry, latitude, fetch, shoreline shape, etc.). [citation] Ice cover in the Great Lakes has been recorded for over five decades with information from high resolution, satellite-based instruments. Additionally,

2 Data & Methods

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References

- Associates, M. (2023). Economic impacts of maritime shipping in the great lakes - st lawrence region. <http://marinedelivers.com>
- Austin, J. A., & Colman, S. M. (2007). Lake superior summer water temperatures are increasing more rapidly than regional air temperatures: A positive ice-albedo feedback. *Geophysical Research Letters*, 34(6). <https://doi.org/10.1029/2006gl029021>